

Project 1 – Task Split & Notes

Feedforward Neural Network with Backpropagation

CS 4033/5033 Machine Learning Fundamentals — Spring 2026

1 Division of Work

Code & Data (done)	<code>ann.py</code> and everything in <code>results/</code> . All 9 datasets have been run with 10 repetitions each.
Write-up (yours)	The report covering the underlined steps in the assignment PDF (Steps 2–5, 7–9, 11, 13, 15, 17, 19, 21, 23, 25, 27).

Everything below is a reference for the write-up — the design decisions that were made, the results, and some notes on what to emphasize.

2 Choices Made by the Professor (Steps 2–3)

These were fixed by the assignment:

Choice	What was decided
Task type	Binary classification (two classes)
Network type	Feedforward neural network
Learning algorithm	Supervised, error backpropagation
Output neurons	1
Datasets	Provided CSVs with known labels
Number of classes	2 (Class 0 and Class 1)

3 Our Design Choices (Steps 4–5, 7–8)

Choice	Option	Why
Activation	Sigmoid (all layers)	Outputs in $[0, 1]$; natural fit for binary classification.
Loss	MSE	Simple and effective with sigmoid outputs.
Weight init	Xavier/Glorot	Scales to fan-in/fan-out, avoids vanishing/exploding gradients.
Learning rate	0.1	Standard starting point for sigmoid networks.
Momentum	0.9	Speeds up convergence, helps with local minima.
Epochs	1000	Enough for convergence (verified by learning curves).
Gaussian 2D arch	$[2, 8, 1]$	Roughly linearly separable; one small hidden layer suffices.
Gaussian 3D arch	$[3, 16, 1]$	Extra input dimension, slightly more capacity.
Moons 2D arch	$[2, 16, 8, 1]$	Nonlinear boundary needs two hidden layers.
Normalization	Z-score	Equal feature contribution; stats from training set only.
Split	60/20/20	Standard train/validation/test split.
Repetitions	10	Gives reliable mean $\pm$ std without being excessive.
Varies per rep	Split seed + weight seed	Controls for both data ordering and initialization variance.

4 Data Collection Plan (Step 9)

Sample counts. Gaussian: 200 total $\rightarrow$ 120 train / 40 val / 40 test. Moons: 1000 total $\rightarrow$ 600 train / 200 val / 200 test.

Repetitions. 10 per dataset. Each rep uses a different random seed for the data split and weight initialization. Architecture, learning rate, momentum, and epochs stay fixed.

Metrics collected. Per epoch: training and validation loss (MSE) and accuracy. Per repetition: final test accuracy. Aggregated: mean and standard deviation across reps.

Reporting format. Summary table (mean $\pm$ std), averaged learning curve plots, decision boundary plots (2D datasets), and a comparison bar chart.

Conclusions we can draw. Effect of class separation (wide/narrow/overlap), effect of dimensionality (2D vs. 3D), effect of data shape (Gaussian vs. moons), and where the network hits the Bayes error floor.

5 Results (Steps 10–27)

Dataset	Architecture	Mean Test Acc	Std
Gaussian 2D Wide	[2, 8, 1]	100.00%	0.00%
Gaussian 2D Narrow	[2, 8, 1]	99.75%	0.75%
Gaussian 2D Overlap	[2, 8, 1]	92.00%	3.12%
Gaussian 3D Wide	[3, 16, 1]	100.00%	0.00%
Gaussian 3D Narrow	[3, 16, 1]	98.75%	1.68%
Gaussian 3D Overlap	[3, 16, 1]	96.00%	2.78%
Moons 2D Wide	[2, 16, 8, 1]	87.55%	1.27%
Moons 2D Narrow	[2, 16, 8, 1]	86.95%	1.29%
Moons 2D Overlap	[2, 16, 8, 1]	86.00%	1.26%

Each dataset needs a short paragraph in the report — accuracy, what the learning curve and decision boundary show, and a takeaway. Key points per dataset:

- **Gaussian 2D Wide** — Perfect accuracy, fast convergence, clean linear boundary. Easiest problem.
- **Gaussian 2D Narrow** — Near-perfect (99.75%). Tighter boundary but still separable.
- **Gaussian 2D Overlap** — Drops to 92%. Some samples are inherently misclassifiable (Bayes error). Higher variance across reps (3.12%) because of sensitivity to which overlapping points end up in the test set.
- **Gaussian 3D Wide** — Perfect. Extra dimension doesn't hurt when classes are well-separated.
- **Gaussian 3D Narrow** — 98.75%. Slightly harder than 3D Wide; occasional misclassification near the boundary.
- **Gaussian 3D Overlap** — 96%. Notably better than 2D Overlap (92%), likely because the extra dimension provides more discriminative information.
- **Moons 2D Wide** — 87.55%. Nonlinear boundary needs the deeper architecture. Decision boundary is curved.
- **Moons 2D Narrow** — 86.95%. Minimal degradation from Wide.
- **Moons 2D Overlap** — 86%. Lowest overall. Overlapping crescents create an unavoidable error zone, but low std (1.26%) shows the network is consistent.

6 Overall Conclusions

Main takeaways for the closing section:

- The FFNN learned all 9 problems successfully, with accuracy scaling predictably with difficulty.

- Class separation is the biggest factor: Wide > Narrow > Overlap across the board.
- Moons data is harder than Gaussian — nonlinear boundaries need deeper networks.
- Overlap datasets show the Bayes error floor; no classifier can do perfectly when distributions overlap.
- Architecture should match complexity: one hidden layer for Gaussian, two for moons.
- 10 repetitions with varied seeds gave stable, reproducible results.

7 Files

For submission we need three things: the write-up, the code (`ann.py`), and the data (`results/` folder). The write-up should include a brief note on how the data is organized:

- `grand_summary.csv` — aggregate results for all datasets
- `*_raw.csv` — per-repetition accuracy (one file per dataset)
- `*_curves.png` — averaged learning curves (one per dataset)
- `*_boundary.png` — decision boundaries (2D datasets only)
- `summary_accuracy.png` — bar chart comparing all datasets